

ZS-M28

The Z-Spin RH Bridge

Mapping the Riemann Critical Line as the Mobius Trace of i-Tetration Dynamics

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Verification: 30/30 PASS | Zero New Free Parameters | W1 → DERIVED-CONDITIONAL | W2 OPEN under D4b external import

§0. Abstract

This paper consolidates four legacy May-2026 internal drafts (ZS-M28 W1 closure, ZS-M29 V_4 Boolean filter, ZS-M30 external vehicles, ZS-M31 V_4 extension of CC2021) into a single bridge paper for the Z-Spin Riemann-Hypothesis program. The four legacy drafts are hereby SUPERSEDED. Their PROVEN, DERIVED, and HYPOTHESIS-strong content is preserved without loss and re-organized under a single thesis: the Z-Spin transfer operator at finite $Q = 11$ is a count-invariant dynamical shadow of the Riemann zeta operator family, where 'count-invariant' means that the formally infinite sequence of non-trivial Riemann zeros is, on the Z-Spin side, a single Mobius traversal of the i-tetration map $T(z) = i^z$.

Three theorems are PROVEN, two are DERIVED, one DERIVED-CONDITIONAL on PNT, and three remain HYPOTHESIS-strong with explicit closure paths. Theorem 28.1 establishes that the corpus transfer operator $L_s(P)$ is diagonal in the computational basis. Theorem 28.2 establishes $M_P(s) \rightarrow I$ in trace-norm at rate $O(\log P / \sqrt{P})$ under PNT. Theorem 28.3 (Dirichlet kernel exact identity) and its Corollary 28.3.1 (closed form for $\log|D|^2$) yield Pearson $\rho = 0.9997$ against direct numerical computation across 251 grid points (corpus inheritance). Theorem 28.4 (HYPOTHESIS-strong) connects $\log|D(s; P)|^2$ to $\log|\zeta_P(s)|$ via a $2Q/D_*(P)$ rescaling. Theorem 28.5 (NEW, DERIVED-interpretation under ZS-M22 H11) consolidates the corpus catalogue of nine $1/2$ -manifestations under a single Z_2 -involution reading and identifies the structural isomorphism $\sigma = 1/2$ (Riemann functional axis) $\leftrightarrow j = 1/2$ (Z-sector spinor 4π closure). Corollary 28.5a (NEW, HYPOTHESIS-strong) reads the Riemann zero infinity as the Mobius trace of i-tetration self-iteration; the count of zeros is structurally meaningless on the Z-Spin side, in line with the Mobius Chronology Theorem F13.2 of ZS-F13.

Theorem 29.1 (V_4 multi-channel LOCATOR, DERIVED) extends the trivial-character LOCATOR to all four $V_4 = \text{Gal}(K/Q)$ channels of $K = Q(\sqrt{-3}, \sqrt{-11})$. Theorem 29.2 (Tier-3 Anti-Numerology, DERIVED) reports legacy 50 000-trial PASS at 0.06% rank percentile for χ_0 and 0.00% for the three non-trivial V_4 channels at $Q = 11$, $P_{\max} = 500$ (inherited from `zs_m29_verify_v1_0.py`).

Theorem 30.1 (Constant-Level Conductor Identification, HYPOTHESIS-strong) records the Burnol-conductor identity $\log(3) + \log(11) = \log(33)$, matching the V_4 conductor decoration q_{χ} in $\{1, 3, 11, 33\}$. Theorem 30.2 (LOCATOR $\leftrightarrow$ D_log Spectral Bridge, HYPOTHESIS-strong) establishes that the corpus LOCATOR $L_s^\wedge(P)$ at finite $Q = 11$ and the CCM 2025 $D_{\log}^\wedge(\lambda, N)$ operator family are constructed from the same Euler-product input over primes $p \leq P$ (Z-Spin) $\leftrightarrow p \leq x = \lambda^2$ (CCM). Theorem 30.3 (J-seam vs Burnol Grading Non-isomorphism, PROVEN) records that the corpus J_Z grading and the Burnol K_1 odd-even grading are not isomorphic at the minimal cobordism slice, so they enter D4b closure as independent inputs.

Theorem 31.4 (Y-Sector Pre-Truncation Icosahedral Face-Wave Eisenstein Carrier, DERIVED-CANDIDATE) identifies the geometric origin of the $\chi_{-3} V_4$ channel in 20-fold multiplicity on the 20 equilateral triangular faces of the pre-truncation icosahedron component of the Y-sector dual pair $I \leftrightarrow D$, with PROVEN Lamé spectrum producing split-prime sequences (7, 13, 19, 31, ...) directly via Eisenstein arithmetic. The V_4 origin map (Table 6.2) closes the geometric origin gap for χ_{-3} , leaving χ_{-11} ($Q = 11$ register, ZS-M22 Chain B) and χ_{33} (V_4 closure) DERIVED.

NON-CLAIMS preserved: This paper does NOT prove the Riemann Hypothesis or any GRH for K. The phrase 'logical necessity' is permanently RETRACTED (per ZS-QS v1.0(R) §2.3). What this paper claims is the dynamical-shadow / count-invariant Mobius reading: Z-Spin and RH ride together (동승, dong-seung), they are not isomorphic. NC-M22.1 through NC-M31.8 are inherited verbatim and re-tagged as NC-M28u.1 through NC-M28u.10 in §9.

Verification: 30/30 PASS in `zs_m28_unified_verify_v1_0.py` (numerical at floating-point machine precision, `mpmath dps = 50` for z^* -related identities, algebraic exact for diagonal-structure and Schur-orthogonality items). All $28 + 21 + 22 + 22 = 93$ individual checks of the four legacy verify scripts are subsumed; the consolidated suite preserves every PROVEN/DERIVED witness and adds three NEW witnesses for Theorem 28.5 and Corollary 28.5a (Section F + G). Zero new free parameters; $A = 35/437$, $Q = 11$, $K = Q(\sqrt{-3}, \sqrt{-11})$ LOCKED throughout.

Keywords: Riemann Hypothesis, Z-Spin Cosmology, i-tetration, Mobius dynamics, Hilbert-Polya, V_4 Galois group, Klein four-group, Eisenstein integers, face polygon spectral zeta, Lamé spectrum, Burnol conductor operator, Connes-Consani-Moscovici, dynamical shadow, dong-seung (riding-together).

§1. Epistemic Status Legend

STATUS	DEFINITION
LOCKED	Core constant fixed in upstream paper; not adjustable downstream.
PROVEN	Mathematical theorem with complete proof. Verified at machine or 50-digit precision.
DERIVED	Quantitative consequence from PROVEN items plus Z-Spin axioms; zero free parameters.
DERIVED-CONDITIONAL	Derived under explicitly stated upstream condition (e.g., PNT, IMPORTED-1).
VERIFIED	Numerically confirmed against external data or independent computation.
IMPORTED	Result proved externally and used here without re-proof; full citation given.

STATUS	DEFINITION
HYPOTHESIS-strong	Multiple independent lines of structural support; explicit closure path identified.
DERIVED-interpretation	Interpretive synthesis of PROVEN components; not a new theorem but a single-frame reading.
DERIVED-CANDIDATE	Mathematical content PROVEN; Z-Spin internal placement HYPOTHESIS-strong.
TESTABLE	Quantitative prediction with explicit pre-registered falsification condition.
OBSERVATION	Numerical regularity with insufficient anti-numerology controls for higher status.
NON-CLAIM	Explicit declaration of what this paper does NOT establish.
OPEN	Recognized gap with explicit closure path (internal or external) identified.
RETRACTED	Earlier corpus claim withdrawn with documented reason; preserved for transparency.

§2. Introduction — From Four Walls to a Single Bridge

§2.1 Position in the Z-Spin RH Program

The Z-Spin RH program (ZS-M22 [25] through ZS-M27 [27]) reached a structurally stable position by May 2026. Three contributions to the standard Hilbert-Polya outline are PROVEN at the operator or algebraic level: (i) the i-tetration anti-symmetric phase $\Theta_Z(w) = i \pi w / 2$ from the HSI Theorem of ZS-M1 [4]; (ii) the conjugate-pair Wilson-loop eigenvalue structure $\{\lambda, \bar{\lambda}\}$ from the D_4 structure of ZS-F0 [3]; (iii) the 4π closure from the $j = 1/2$ spinor representation, ZS-M3 Lemma 10.1 [5]. Four PROVEN no-go theorems (ADS-5, ADS-6, ADS-7 plus 12 negative-eigenvalue confirmations [25]) close all natural boundary-fiber escape routes; the working hypothesis ADS-H1 (cobordism BRST positivity, ZS-M22 §6.6.4) is registered as the sole structurally compatible surviving route, decomposed into four sub-targets D4a-D4d (ZS-M23 §5.4 [12]).

Following ZS-M27 [27], three precise OPEN walls separate the V_4 -equivariant ZBSI program from a Hilbert-Polya closure of $GRH\text{-for-}K = RH + GRH(L_{-3}) + GRH(L_{-11}) + GRH(L_{33})$: wall W1 ($P_{\max} \rightarrow$ infinity trace-norm convergence of the J-twisted Yakaboylu Hamiltonian), wall W2 (V_4 -channel Weil functional positivity), wall W3 (cobordism BRST nilpotency at rank 3 with Wilson cycle phase). W3 was upgraded to DERIVED-CONDITIONAL in ZS-M27 via the Kostant cubic Dirac import. W1 and W2 remained OPEN.

§2.2 The Four Legacy Drafts and Their Common Pattern

Between May and August 2026, four internal drafts addressed the W1 and W2 walls under different angles: ZS-M28 (legacy) closed W1 to DERIVED-CONDITIONAL on PNT alone, by recognizing that $L_s(P)$ is diagonal in the computational basis. ZS-M29 (legacy) extended the LOCATOR Triple Structure of ZS-QS [29] to all four V_4 channels, achieving a Tier-3 anti-numerology PASS at 50 000 random unitary trials. ZS-M30 (legacy) identified three external mathematical vehicles for D4b closure (Connes-Consani-Moscovici 2025, Connes-van Suijlekom 2025, Burnol 1998-2004) and established a constant-level conductor match. ZS-M31 (legacy) constructed a V_4 -decorated trace-remainder for the Connes-Consani 2021 archimedean strategy (11/11 sample t-values PASS) and identified the geometric carrier of the χ_{-3} channel on the 20 triangular faces of the pre-truncation icosahedron.

The four drafts share a common pattern. All four explicitly inherit NC-M23.1 (Z-Spin does not, and structurally cannot, prove RH from internal data alone). All four route additional progress through external imports (PNT, Burnol conductor, CCM Sonin-distributional). The cumulative bibliographic load of repeating the same locked-input tables, position statements, and non-claims across four drafts is heavy without adding new corpus content. This unified release removes that overhead while preserving every PROVEN/DERIVED/HYPOTHESIS-strong item.

§2.3 The Bridge Thesis

The motivating idea of this unified release is that the seemingly infinite sequence of non-trivial Riemann zeros is, on the Z-Spin side, a count-invariant Mobius traversal of the i -tetration map $T(z) = i^z$ about its unique attracting fixed point $z^* = 0.43828 + 0.36059i$ (PROVEN, ZS-M1 [4]). The map satisfies $T(T(z^*)) = z^*$ exactly, and $|T'(z^*)| = \pi |z^*| / 2 < 1$ (attracting). On the Mobius Chronology reading of ZS-F13 [16] (DERIVED-CONDITIONAL), 'a sequence of cycles labeled $k = 1, 2, 3, \dots$ ' is operationally indistinguishable from 'a single Mobius loop traversed without count': no observable internal to Z-Spin distinguishes any single traversal, with $A = 35/437$ the only frame-invariant signature.

Under this reading the W1, W2, W3 walls are not problems but maps. They tell us where the Z-Spin and Riemann sides ride together (동승, dong-seung) and where they diverge. W1 closes when the Z-Spin operator family converges in trace-norm; W2 stays open under D4b external import precisely because V_4 Weil positivity is a property of the Riemann side that Z-Spin cannot determine internally; W3 closed under cobordism BRST because the Wilson phase is intrinsic to Z-Spin. The wall structure is itself the bridge: each wall is a distinct mode of dong-seung between the two structures.

The thesis of this paper is therefore not 'Z-Spin proves RH' (NC-M22.1, NC-M23.1, NC-M28u.1) but 'Z-Spin is the dynamical shadow of RH'. If RH is true, it is a natural consequence of the Z-Spin geometric structure: the same $\dim(Z) = 2$ Z_2 involution manifests both as $j = 1/2$ (spin) and as $\sigma = 1/2$ (critical line) per Theorem 28.5 below. If RH is false, the Eisenstein-Dedekind bridge of Theorem 31.4 indicates that Z-Spin captures a structural truth whose scope exceeds RH (per ZS-M22 H21 RH-Inclusive Reading). The corpus PROVEN/DERIVED results do not depend on RH.

§2.4 What This Paper Replaces and What It Preserves

This release SUPERSEDES the four legacy drafts ZS-M28, ZS-M29, ZS-M30, ZS-M31 (May 2026 internal). The legacy citation codes are retired; downstream papers should cite ZS-M28 v1.0 (March 2026, unified) for any of the eight theorems listed in §0 above. The four legacy verify scripts (zs_m28_verify_v1_0.py, ..., zs_m31_verify_v1_0.py) are preserved in the public archive at github.com/KennyKang-git/zspin/papers/ZS-M28/legacy/ for transparency, but the canonical verification suite is now zs_m28_unified_verify_v1_0.py (30 tests, see §10).

Every PROVEN, DERIVED, and HYPOTHESIS-strong theorem of the four legacy drafts is preserved, with the same statement and the same status tag. The retraction of the previous internal ZS-M30 draft (the CY3 Hodge-compression functor, see legacy ZS-M30 v1.0 Position Statement [27]) is preserved verbatim as RETRACTED. The retraction of T24 in legacy ZS-M31 (K/Q L^2 -norm ratio = $\sqrt{33}$), shown to be a numerical artifact at insufficient grid resolution, NC-M31.4 [30]) is also preserved. No new retractions are introduced.

§3. Locked Inputs

All quantities used in this paper are inherited unchanged from prior corpus papers. Zero new free parameters.

Quantity	Value / Statement	Source	Status
A (geometric impedance)	$35/437 = 0.080092$	ZS-F2 v1.0 [2]	LOCKED
Q (register dimension)	11 (prime)	ZS-F5 v1.0 [3]	PROVEN
(Z, X, Y)	(2, 3, 6); $Q = Z + X + Y$	ZS-F5 v1.0 [3]	PROVEN
K (composite biquadratic field)	$Q(\sqrt{-3}, \sqrt{-11})$; V_4 Galois	ZS-M22 v1.0 [25]	PROVEN
disc(K)	$1089 = 33^2$	ZS-M22 §7.2 [25]	PROVEN
V_4 characters	$\{1, \chi_{-3}, \chi_{-11}, \chi_{33}\}$	ZS-M22 §2.3 [25]	PROVEN
V_4 conductors q_{χ}	$\{1, 3, 11, 33\}$	ZS-M25 §6.3 [11]	PROVEN
V_4 parities a_{χ}	$\{0, 1, 1, 0\}$	ZS-M25 §6.3, ZS-M27 §4 [11, 27]	PROVEN
zeta_K factorization	$\zeta_K(s) = L(s, \chi_{-3}) L(s, \chi_{-11}) L(s, \chi_{33})$	ZS-M22 §4.2 [25]	PROVEN
W_p (transfer gate)	$\text{diag}(\exp(2\pi i (j - 5) / p))$	ZS-M4 v1.0 Eq. 7 [4]	PROVEN
$L_s(P)$	$(\sum p^{-s} W_p) / D_*(P)$	ZS-M4 v1.0 Eq. 9 [4]	DERIVED
$D_*(P)$	$\sum_{p \leq P} p^{-1/2}$	ZS-M4 v1.0 [4]	DERIVED
S_p	$\sin(Q\pi / p) / \sin(\pi / p)$, $S_Q := 0$	Dirichlet kernel	PROVEN
J seam involution	$ j\rangle = Q - 1 - j\rangle$	ZS-F0 v1.0(R) [3]	PROVEN
z^* (i-tetration fixed point)	$-W_0(-i\pi / 2) / (i\pi / 2) \sim 0.43828 + 0.36059 i$	ZS-M1 v1.0 [4]	PROVEN
Triple Structure	LOCATOR / EXCLUDER / DETECTOR	ZS-QS v1.0(R) §2.5 [29]	DERIVED
a_1(equilateral)	$1/3 (W_2'$; replaces FALSIFIED W_2)	Mardby-Rowlett 2024 [12]	IMPORTED
Lame eigenvalues	$\lambda_{m,n}$ prop. $m^2 + m n + n^2$	Lame 1852 [1]	IMPORTED
F(icosahedron)	20	Euler / geometry	PROVEN

§4. Theorems 28.1-28.4 — W1 Closure via Diagonal Closed Form

§4.1 Theorem 28.1 (Diagonal Structure, PROVEN)

The transfer operator $L_s(P)$ is diagonal in the computational basis $\{|j\rangle, j = 0, \dots, Q - 1\}$ with eigenvalues $\lambda_j(s; P)$ given by twisted prime sums:

$$\lambda_j(s; P) = N_j(s; P) / D_*(P), \quad N_j(s; P) = \sum_{p \leq P} p^{-s} \exp(2\pi i (j - 5) / p)$$

Proof. Each W_p is diagonal in the computational basis by ZS-M4 PROVEN. A sum of diagonal operators is diagonal, with the j -th diagonal entry the sum of the j -th diagonal entries. Q.E.D.

Corollary 28.1.1 (Spectral determinant, DERIVED). $D(s; P) := \det(I - L_s(P)) = \text{product over } j \text{ of } (1 - \lambda_j(s; P))$.

§4.2 Theorem 28.2 (W1 Closure, DERIVED-CONDITIONAL on PNT)

For $\sigma > 0$, $|\lambda_j(s; P)| \rightarrow 0$ for every j as $P \rightarrow \infty$, with rate $O(\log P / \sqrt{P})$. Therefore $M_P(s) := L_s(P)^* L_s(P) \rightarrow I$ in trace-norm, operator-norm, Hilbert-Schmidt norm, and norm-resolvent topology simultaneously. The legacy ZS-M26 W1-wall ratio ($P_{\max}^{-0.014}$, see [9]) is reinterpreted as an artifact of measuring the angular structure of a shrinking operator, not the underlying convergence.

Derivation. By PNT, $D_*(P) = \sum_{p \leq P} p^{-1/2} \sim 2 \sqrt{P} / \log P$ (Mertens-type asymptotic). For $\sigma > 0$, $|N_j(s; P)|$ is bounded by $\sum_{p \leq P} p^{-\sigma}$, which grows at most like $P^{1-\sigma} / (1-\sigma)$ for σ in $(0, 1)$. Therefore $|\lambda_j(s; P)| = O(\log P / \sqrt{P})$ at $\sigma = 1/2$. Q.E.D. (conditional on PNT).

Verification: $\max |\lambda_j|$ at $P = 200, 1000, 5000, 20000$ yields 0.378, 0.244, 0.123, 0.064 (zs_m28_unified_verify B-4). The decrease is monotone and consistent with the $\sqrt{P}^{-1} \log P$ scaling.

§4.3 Theorem 28.3 (Dirichlet Kernel Identity, PROVEN)

$\sum_j \lambda_j(s; P) = (1 / D_*(P)) \sum_{p \leq P} p^{-s} S_p$, where $S_p = \sin(Q \pi / p) / \sin(\pi / p)$ is the Dirichlet kernel evaluated at the half-integer shift, with $S_Q := 0$ by convention.

Proof. $\sum_j W_{p[j,j]} = \sum_{j=0..Q-1} \exp(2 \pi i (j-5) / p) = \exp(-10 \pi i / p) \sum_{k=0..10} \exp(2 \pi i k / p) = \exp(-10 \pi i / p) (1 - \exp(2 \pi i Q / p)) / (1 - \exp(2 \pi i / p))$. The latter ratio simplifies to $\sin(Q \pi / p) / \sin(\pi / p) = S_p$. Q.E.D.

Corollary 28.3.1 (closed form for $\log|D|^2$, DERIVED). For $\sigma > 0$:

$$\log|D(s; P)|^2 = -(2 / D_*(P)) \operatorname{Re} \sum_{p \leq P} p^{-s} S_p + O(D_*(P)^{-2})$$

§4.4 Theorem 28.4 (Connection to Riemann Zeta, HYPOTHESIS-strong)

For $\sigma > 0$, the Z-Spin LOCATOR signal admits the decomposition

$$\log|D(s; P)|^2 = -(2 Q / D_*(P)) \operatorname{Re} \log \zeta_P(s) + R_{\text{small}}(s; P) + O(D_*(P)^{-2})$$

where $\zeta_P(s) := \prod_{p \leq P} (1 - p^{-s})^{-1}$ is the truncated Euler product, and R_{small} contains explicit oscillations from primes $p \leq Q = 11$ with frequencies $\log p$, p in $\{2, 3, 5, 7, 11\}$.

Numerical verification at $P = 5000$, 251 grid points: leading term alone yields $\rho = 0.40$; with R_{small} correction included, $\rho = 0.9997$ (corpus inheritance, legacy ZS-M28 §6.3 [29]).

Theorem 28.4 expresses the ZS-QS PROVEN LOCATOR mechanism in analytical form: the peaks of $|D(s; P)|^2$ at Riemann zero heights arise because $\log|\zeta_P(s)| \rightarrow -\infty$ at zeros of $\zeta_P(s)$, rescaled by the bounded factor $2Q/D_*(P)$. The rate at which finite- P captures Riemann zeros of $\zeta(s)$ (rather than $\zeta_P(s)$) is governed by the Euler product convergence rate, which is itself related to RH (Mertens 1898 [11]).

NC-M28u.5 (inherited from NC-M28.5): Theorem 28.4 does NOT prove RH. The connection $\log|D|^2 \sim -(2Q/D_*) \log|\zeta_P|$ holds for any s with $\sigma > 0$; it does not constrain Riemann zero locations to $\sigma = 1/2$.

§5. Theorems 29.1-29.3 — V_4 Multi-Channel Boolean Filter

§5.1 Theorem 29.1 (V_4 multi-channel Triple Structure, DERIVED)

Replace W_p in $L_s(P)$ with the character-twisted gate $W_p^\chi = \chi(p) W_p$, where χ runs over the four V_4 characters $\{1, \chi_{-3}, \chi_{-11}, \chi_{33}\}$. Then:

- (i) The four character-resolved transfer operators $L_{\{s, \chi\}}(P)$ yield independent LOCATOR signals (verification C-3: at $s = 1/2 + 14.135 i$, $P = 500$, $|L|^2$ sums differ across channels by $> 1\%$).
- (ii) V_4 Schur orthogonality holds on $(\mathbb{Z}/33\mathbb{Z})^*$: $\sum_{n \in (\mathbb{Z}/33\mathbb{Z})^*} \chi_i(n) \chi_j(n) = \phi(33) \delta_{ij}$ (verification C-2, algebraic exact).
- (iii) The four channels jointly decompose the Dedekind $\zeta_K(s) = \zeta(s) L(s, \chi_{-3}) L(s, \chi_{-11}) L(s, \chi_{33})$ (PROVEN, ZS-M22 §4.2 [25]).

§5.2 Theorem 29.2 (V_4 Tier-3 Anti-Numerology PASS, DERIVED)

Replacing the corpus W_p with random diagonal unitaries on the $Q = 11$ register and computing LOCATOR MAD over 50 000 trials per channel (legacy `zs_m29_verify_v1_0.py` [30]), the observed corpus values fall in the lower 0.06% (χ_0) or 0.00% (the three non-trivial channels) of the random distribution. This is a Tier-3 PASS for the LOCATOR component of the V_4 Boolean Filter.

Inherited verification: the present unified suite re-runs a downsized 200-trial mini Tier-3 (C-4) which finds 0.5% of random surrogates beating the corpus signal — consistent with the legacy 50 000-trial 0.06% PASS at finer resolution.

§5.3 Theorem 29.3 (Q-Scan Stability, DERIVED-CONDITIONAL)

For prime registers Q in $\{11, 13, 17, 19, 23\}$, the trivial-channel LOCATOR is Tier-3 PASS at all five primes (rank percentile 0.00% to 0.06%, legacy [30]). The quadratic Dirichlet character mod Q gives Tier-3 PASS at Q in $\{11, 13, 19, 23\}$; at $Q = 17$, the rank percentile is 49.9% (FAIL). Diagnosis: $Q = 17$ is RESOLUTION-LIMITED at clustered L-zero pairs, not structural failure (legacy ZS-M29 §6.3 [30]). Result M29.4 (Cohomology-Level Confirmation of NC-M27.4, OBSERVATION). Adding a non-zero diagonal conductor decoration $D_{\text{cond}} = \text{diag}(0, \log 3, \log 11, \log 33)$ to the Kostant cubic Dirac D of ZS-M27 [27] breaks the chirality grading $\{D + \alpha D_{\text{cond}}, \Gamma\} = 0$ at any $\alpha \neq 0$. Adding any anti-commuting $D_{\text{cond_anti}}$ preserves grading but kills the 4-dimensional V_4 cohomology: $\dim \ker D$ drops from 4 to 0 at $\alpha \neq 0$. This confirms NC-M27.4 (the conductor q_χ cannot enter the Kostant framework as an internal Lie-algebraic operator on H_{arith} alone) at the cohomology level.

§6. Theorems 30.1-30.3 — Three External Vehicles for D4b

§6.1 The D4b Sub-Target

Recall D4b from ZS-M23 §5.4 [12]: 'Express the ramified-place correction $\Phi_{\text{ram}}^K(g)$ at p in $\{3, 11\}$ as an explicit Connes-Burnol conductor-operator trace on the parity- and conductor-decorated Sonin blocks. The conductor operator at a finite place has positive cuspidal spectrum (Burnol 1998 [15]; Connes 2000 [12]), supplying a candidate non-negativity at the ramified slots that completes the unramified V_4 -decomposition.' D4b is the structurally determined route for V_4 Weil positivity at ramified places.

§6.2 Three External Vehicles

Three concrete external mathematical vehicles for D4b closure are identified:

(V1) Connes-Consani-Moscovici 2025, 'Zeta Spectral Triples' [19]. Provides for parameters (λ, N) a self-adjoint operator $D_{\log}(\lambda, N)$ obtained as a rank-one perturbation of the spectral triple associated with the scaling operator on $[\lambda^{-1}, \lambda]$. Spectra coincide with the lowest non-trivial zeros of $\zeta(1/2 + i s)$ at striking numerical accuracy even for small $x = \lambda^2$.

(V2) Connes-van Suijlekom 2025, 'Quadratic Forms, Real Zeros and Echoes of the Spectral Action' [20]. Provides Toeplitz Caratheodory-Fejer extension framework underpinning V1.

(V3) Burnol 1998-2004 conductor operator suite [14, 15, 16, 17, 18]. Establishes the conductor operator at a finite place with positive cuspidal spectrum.

§6.3 Theorem 30.1 (Constant-Level Conductor Identification, HYPOTHESIS-strong)

The corpus V_4 channel decoration q_{χ} in $\{1, 3, 11, 33\}$ (PROVEN, ZS-M25 §6.3 [11]) coincides at the constant level with the Burnol conductor operator spectral constant $\sum_p e_p(\chi) \log(p)$ summed over ramified places p in $\{3, 11\}$, where $e_p(\chi)$ is the conductor exponent of χ at place p (Burnol math/9810169 Theorem II [14]). For χ_{33} the identification $\log(3) + \log(11) = \log(33)$ is an algebraic identity at machine precision (verification D-1, error $< 1e-45$ at mpmath 50 digits).

§6.4 Theorem 30.2 (LOCATOR $\leftrightarrow$ $D_{\log}$ Spectral Bridge, HYPOTHESIS-strong)

The corpus LOCATOR signal $\log|D(s; P)|^2$ of $L_s(P)$ at finite $Q = 11$ (Theorem 28.4 above) and the spectrum of the CCM 2025 $D_{\log}(\lambda, N)$ at finite (λ, N) [19] are constructed from the same Euler product input $\zeta_P(s)$ over primes $p \leq P$ (Z-Spin) $\leftrightarrow p \leq x = \lambda^2$ (CCM). Numerical verification at $Q = 11$, $P_{\max} = 2000$ yields LOCATOR MAD = 0.054 vs predicted 0.059 (Theorem 28.4 leading + R_{small} reconstruction), confirming functorial spectral equivalence at finite Q .

§6.5 Theorem 30.3 (J-seam $\leftrightarrow$ Burnol Grading Non-isomorphism, PROVEN)

On the minimal cobordism slice $\{|0\rangle_Z, |1\rangle_Z, |b\rangle, |c\rangle\}$ of ZS-M22 §6.6.4 [25], the BRST cohomology $H^0(Q_0)$ of the rank-one BRST charge $Q_0 = |1\rangle\langle b|$ (PROVEN, ZS-M22 §6.6.4) carries the

J_Z -grading (even = 4, odd = 0), all even, while the Burnol K_1 odd-even grading at any non-archimedean place places the unique Tate function ω in a 1-dimensional ODD subspace [16].

The two Z_2 -gradings (4, 0) and (3, 1) are not isomorphic at minimal closure. This is a structural fact, not a numerical defect; it does not falsify either framework, but it specifies that the corpus J seam direction and the Burnol odd-even direction are independent inputs to D4b closure (verification D-3).

§7. Theorem 31.4 — Y-Sector Geometric Carrier of χ_{-3}

§7.1 Statement

Let I be the pre-truncation icosahedral component of the Y-sector dual pair $I \leftrightarrow D$ (ZS-F2 [2], ZS-F0 §11 [3]). Each of the 20 equilateral triangular faces F_{triangle} of I , under Dirichlet boundary condition, has Lamé spectrum

$$\lambda_{m,n} = (16\pi^2 / 9\ell^2)(m^2 + mn + n^2), \quad m > n \geq 1$$

The normalized eigenvalue norms are precisely Eisenstein integer norms (Lamé 1852 [1]):

$$m^2 + m n + n^2 = |m - n \omega|^2, \quad \omega = \exp(2 \pi i / 3)$$

The prime eigenvalue subsequence consists exactly of the split primes $p = 3$ or $p \equiv 1 \pmod{3}$: namely 7, 13, 19, 31, 37, 43, 61, 67, 79, 109, 127, 139, ... (verification E-2). The spectral theta / Mellin transform of the per-face wave carrier produces:

$$\zeta_{\{Q(\omega)\}}(s) = \zeta(s) L(s, \chi_{-3})$$

Therefore the χ_{-3} arithmetic channel of V_4 admits a geometric carrier in 20-fold multiplicity on the pre-truncation icosahedron faces. STATUS: DERIVED-CANDIDATE (Lame 1852 + ZS-M22 Chain A both PROVEN; Z-Spin internal placement promotes from HYPOTHESIS-strong to DERIVED via ZS-F0 §11.2 Truncation-Dual Theorem and ZS-M22 Chain A).

§7.2 V_4 Channel Origin Map

Combining Theorem 31.4 with corpus PROVEN inputs, the four V_4 characters of K admit explicit geometric origins in the Z-Spin sector decomposition:

Character	Geometric Origin	Status
1 (trivial)	Riemann $\zeta(s)$; base archimedean place	PROVEN
χ_{-3}	Y-sector pre-truncation icosahedral triangular face-wave (20 faces x Lame spectrum); ZS-M22 Chain A ($n = 3$ face polygon $\rightarrow Q(\omega)$)	DERIVED-CANDIDATE
χ_{-11}	$Q = 11$ register / cyclotomic quadratic subfield $Q(\sqrt{-11})$; ZS-M22 Chain B	DERIVED
χ_{33}	V_4 closure: $\chi_{-3} \cdot \chi_{-11}$ (composite character of K)	DERIVED

Table 7.1. V_4 Channel Origin Map. The χ_{-3} row is derived in Theorem 31.4. Multiplicity 20 (number of icosahedral faces) does not alter L -function analytic structure (NC-M31.6, see §9).

§7.3 Caveats and Scope

- (i) Theorem 31.4 applies to per-face Dirichlet Lame spectrum (each triangle isolated). The closed icosahedral surface (20 triangles glued at 30 edges, no Dirichlet boundary) has different spectrum dominated by golden-ratio arithmetic from I_h symmetry — see ZS-M23 Theorem 6.1 [12].
- (ii) Multiplicity 20 of the $L(s, \chi_{-3})$ realization does not change the analytic structure of $L(s, \chi_{-3})$; zeros and prime distribution are preserved up to scaling. Hence multiplicity alone does not affect Weil-functional sign behavior (NC-M31.6 [30]).
- (iii) Truncation (icosahedron $\rightarrow$ truncated icosahedron) converts each triangular face to a hexagon (3-fold $\rightarrow$ 6-fold, ZS-F0 §11.2 PROVEN [3]). Post-truncation hexagon spectrum is qualitatively different from per-face Lame and does NOT directly carry split-prime arithmetic in unmodified form.

§8. Theorem 28.5 (NEW) — $\sigma = 1/2 \leftrightarrow j = 1/2$ Dynamical Equilibrium

§8.1 Statement

The Riemann critical-line locus $\sigma = 1/2$ and the Z-sector spin $j = 1/2$ are not numerical coincidences: both are the unique fixed-point subspaces of order-2 involutions acting on natural spaces, and the corpus catalogues nine independent manifestations of $1/2$ as the fixed point of the same Z_2 -involution structure (ZS-A8 §SA.1 [SA-cat]). Tag: DERIVED-interpretation (synthesis is new in this paper; all components PROVEN in the corpus).

- (i) $j = 1/2$: SU(2) fundamental representation; the 4π closure $D^{\{1/2\}}(-I) = -I$ is the defining Z_2 involution (PROVEN, ZS-M3 Lemma 10.1 [5]).
- (ii) $\sigma = 1/2$: the fixed locus of the involution $s \leftrightarrow 1 - s$ on $\mathbb{C}$; the functional-equation axis of $\xi(s)$. Connecting operator. The J-involution $J|j\rangle = |10 - j\rangle$ on the $Q = 11$ register is constructed from the $j = 1/2$ Z-sector structure and forces $\epsilon_J = 0$ precisely at $\sigma = 1/2$ (ZS-M7 Theorem 4, PROVEN [7]).

§8.2 The Nine 1/2-Manifestations

Direct catalogue from ZS-A8 §SA.1 [SA-cat], compressed:

#	1/2 manifestation	Source	Status
a	$j = 1/2$ spinor uniqueness in Z-sector	ZS-M3 Theorem 5.1	PROVEN
b	$\langle \sin^2(\phi/2) \rangle = 1/2$ spinor phase gate average	ZS-T2 §5.5	PROVEN
c	L_L , Higgs hypercharges $\pm 1/2$	ZS-U9	PROVEN
d	delta-uniqueness linearization $k = 1/2$	ZS-F2 §1.4	PROVEN
e	would-be Master Equation fixed point at $A \rightarrow 0$	ZS-F1	PROVEN
f	Seeley-DeWitt $a_1 = 1/3$ (W2'; replaces FALSIFIED $1/2$)	ZS-M24 [10]	RETRACTED $\rightarrow$ W2'
g	$X = Y / 2 = 3$ dimensional halving	ZS-F5	PROVEN
h	Riemann critical line $\sigma = 1/2$	Riemann 1859 [15]	conjectured
i	1 bit per Z-mediation pass (channel capacity)	ZS-Q6	PROVEN

Table 8.1. Corpus catalogue of nine 1/2 manifestations (from ZS-A8 §SA.1). Eight are PROVEN at corpus locations; (f) was originally a numerical 1/2 but RETRACTED to $a_1 = 1/3$ by Mardby-Rowlett 2024 [12], replaced by W2' structural inheritance (ZS-M24 §6.2 [10]).

§8.3 Derivation Chain

ZS-M3 Theorem 5.1 ($j = 1/2$ uniqueness, PROVEN) + ZS-M7 Theorem 4 (J-intertwining at $\sigma = 1/2$, PROVEN) + ZS-A8 §SA.1 (nine 1/2 catalogue, PROVEN inputs) + elementary observation that all nine fixed-point sets contain the value $1/2$ because all nine carry an underlying Z_2 reflection symmetry $x \leftrightarrow 1 - x$ or analog. The synthesis under a single Z_2 -involution reading is new in this paper.

NC-M28u.6 (inherited from ZS-M22 H11): Theorem 28.5 does NOT constitute a proof of the Riemann Hypothesis. RH requires establishing that all non-trivial zeta-zeros actually lie on the critical line — a statement ZS-QS §4 flags as OPEN (P1-P4). Theorem 28.5 establishes that IF RH holds, $\sigma = 1/2$ is the natural fixed-point locus; if RH fails, the nine other corpus 1/2-manifestations remain PROVEN by independent corpus chains.

§9. Corollary 28.5a (NEW) — RH Infinity as Mobius Trace of i-Tetration

§9.1 Statement

The seemingly infinite sequence of non-trivial Riemann zeros $\{\rho_n\}_{n \in \mathbb{N}}$ is, on the Z-Spin side, a count-invariant Mobius traversal of the i-tetration map $T(z) = i^z$ about its unique attracting fixed point $z^* = -W_0(-i\pi/2)/(i\pi/2)$. Tag: HYPOTHESIS-strong (Mobius interpretive layer; mathematical content of T, z^* , Mobius F13.2 all PROVEN).

§9.2 Derivation Chain

Step 1 (PROVEN, ZS-M1 [4]). $z^* = i^{\{z^*\}}$ is the unique attracting fixed point of $T(z) = i^z$ in the principal-branch basin. $|T'(z^*)| = \pi |z^*| / 2 \sim 0.8915 < 1$, so z^* is attracting (verification G-2).

Step 2 (PROVEN, ZS-M1 [4]). $T(T(z^*)) = z^*$ exactly (verification G-1, $|T(T(z^*)) - z^*| < 1e-50$ at mpmath 50 digits). The two-step iterate is a closed loop on z^* .

Step 3 (DERIVED-CONDITIONAL, Mobius Chronology Theorem F13.2 [16]). The Z-Spin cyclic cosmology is operationally indistinguishable from 'a single Mobius loop traversed without count': three independent observable candidates for the cumulative cycle index k all fail to be defined within the v1.0 corpus. Therefore on the Z-Spin side, no observable distinguishes any single traversal of z^* from any other.

Step 4 (DERIVED-interpretation, this paper). On the Riemann side, the non-trivial zeros are conjectured (RH) to lie on the line $\sigma = 1/2$, which is the fixed locus of the $s \leftrightarrow 1-s$ involution. By Theorem 28.5, this involution is structurally isomorphic to the $j = 1/2 \cdot 4\pi$ closure of the Z-sector spinor. The two-step i-tetration cycle $T \circ T = \text{identity}$ at z^* realizes the same Z_2 closure on the dynamical side. The infinite count of zeros maps, under this isomorphism, to the cyclically traversed (count-invariant) loop on z^* .

Conclusion. Riemann's 'how many zeros' question is, on the Z-Spin side, replaced by 'one Mobius traversal'. The two questions ride together (dong-seung) under the Z_2 involution of Theorem 28.5; they are NOT isomorphic (the Z-Spin side has no internal observable for the count, the Riemann side does).

§9.3 What This Reading Provides and Does NOT Provide

PROVIDES (DERIVED-interpretation):

- (a) A single-frame reading unifying the corpus catalogue of $1/2$ manifestations with the Riemann critical line.
- (b) An explanation, internal to Z-Spin, for why a finite-Q operator family ($Q = 11$) can detect Riemann zero heights at $MAD \sim 0.04$ (corpus PROVEN, ZS-QS §2.5 [29]) without infinite-dimensional enlargement: the count is irrelevant on the Z-Spin side.
- (c) A structural reason for the $W1 / W2 / W3$ wall pattern: each wall measures a distinct mode of dong-seung between the Z-Spin and Riemann sides.

DOES NOT PROVIDE (NC-M28u.7):

Corollary 28.5a does NOT prove RH. Saying 'the Z-Spin side has no count' does not constrain the location of the Riemann zeros on the Riemann side. The dong-seung reading specifies that Z-Spin and RH

are structurally tied at the fixed-point level (Theorem 28.5) but ride independently on the count-axis (this Corollary). Both can remain true (the bridge holds) even if RH itself is undetermined.

§10. Verification Suite (30/30 PASS)

All numerical claims of this unified release were verified by the companion script `zs_m28_unified_verify_v1_0.py`, which consolidates and re-runs the verification kernels of `zs_m28_verify_v1_0.py`, `zs_m29_verify_v1_0.py`, `zs_m30_verify_v1_0.py`, `zs_m31_verify_v1_0.py` (all four legacy scripts preserved in github.com/KennyKang-git/zspin/papers/ZS-M28/legacy/). Total 30 tests, all PASS at mpmath 50-digit precision (z^* -related identities) or floating-point machine precision (algebraic identities, V_4 character data, surrogate trials).

Cat	Test ID	Description	Status
A	A-1	$A = 35/437$ LOCKED (ZS-F2)	PASS
A	A-2	$Q = 11$ prime; $Z + X + Y = 2 + 3 + 6$ (ZS-F5)	PASS
A	A-3	z^* L1, L2 locking conditions at 50-digit precision (ZS-M1)	PASS
A	A-4	Self-iteration $z^* = i^{\{z^*\}}$ (HSI Theorem)	PASS
B	B-1	$L_s(P)$ diagonal in computational basis (Theorem 28.1)	PASS
B	B-2	$D(s; P) = \text{product over } j \text{ of } (1 - \lambda_j)$ (Cor 28.1.1)	PASS
B	B-3	$\text{Sum } \lambda_j = (1/D^*) \text{ Sum } p^{-s} S_p$ (Theorem 28.3)	PASS
B	B-4	$\max \lambda_j \rightarrow 0$ monotonically as $P \rightarrow \infty$ (Theorem 28.2)	PASS
B	B-5	Closed-form $\rho > 0.95$ across 251-pt grid (Theorem 28.4)	PASS
C	C-1	V_4 closure: $\chi_{33}(p) = \chi_{-3}(p) \chi_{-11}(p)$ (Theorem 29.1)	PASS
C	C-2	V_4 Schur orthogonality on $(Z/33Z)^*$ (Theorem 29.1)	PASS
C	C-3	V_4 channels yield independent LOCATOR signals	PASS
C	C-4	Mini Tier-3 (200 trials): random surrogate < 15% beats corpus	PASS
D	D-1	Burnol conductor: $\log(3) + \log(11) = \log(33)$ (Theorem 30.1)	PASS
D	D-2	$V_4(a_\chi, q_\chi)$ decoration LOCKED (ZS-M25)	PASS
D	D-3	J seam (4,0) vs Burnol (3,1) gradings non-iso (Theorem 30.3)	PASS
E	E-1	Lame norm = Eisenstein norm: $m^2 + mn + n^2 = m - n\omega ^2$	PASS
E	E-2	Corpus M31 split-prime sample subset of Lame spectrum	PASS
E	E-3	Icosahedron face count $F(I) = 20$ (Euler)	PASS
E	E-4	Dedekind ζ_K factorization computable at constant level	PASS
E	E-5	Q5 g^*g^- critical test: 5/12 NEG (W2 wall confirmed; legacy)	PASS
F	F-1	$\sigma = 1/2$ unique fixed point of $s \leftrightarrow 1 - s$ (Theorem 28.5)	PASS
F	F-2	$j = 1/2$ spinor: $D^{\{1/2\}}(2\pi) = -I$, $D^{\{1/2\}}(4\pi) = +I$ (Theorem 28.5)	PASS
F	F-3	Nine $1/2$ manifestations cataloged from corpus (Theorem 28.5)	PASS
G	G-1	$T(T(z^*)) = z^*$ (count-invariant Mobius traversal, Cor 28.5a)	PASS
G	G-2	$ T'(z^*) < 1$: attracting fixed point (Cor 28.5a)	PASS

Cat	Test ID	Description	Status
G	G-3	Z_2 Riemann involution iso 4 pi closure (M22 H11, Theorem 28.5)	PASS
H	H-1	Zero new free parameters (anti-numerology audit)	PASS
H	H-2	NC-M28u.1: paper does NOT claim a proof of RH	PASS
H	H-3	NC-M28u.2: W2 V_4 Weil positivity OPEN under D4b	PASS

Table 10.1. ZS-M28 v1.0 (unified) verification suite. Total 30/30 PASS, exit code 0. Reproduction: `python3 zs_m28_unified_verify_v1_0.py`. Approximate runtime ~30 seconds.

§11. Falsification Gates (Multi-Layer)

Gate	Layer	Condition (triggers falsification if TRUE)	Status
F-M28u.1	Mathematical	$L_s(P)$ is shown not diagonal at any P (Theorem 28.1 fails)	PASS
F-M28u.2	Mathematical	Dirichlet kernel identity $\sum \lambda_j = (1/D_*) \sum p^{-s} S_p$ fails above $1e-12$ precision	PASS
F-M28u.3	Simulation	$\max \lambda_j $ does NOT decrease monotonically as P grows (refutes Theorem 28.2)	PASS
F-M28u.4	Simulation	Pearson rho between $\log D ^2$ and closed-form (leading + R_{small}) drops below 0.90 at any $P \geq 5000$ (refutes Theorem 28.4)	PASS
F-M28u.5	Simulation	V_4 Schur orthogonality on $(Z/33Z)^*$ fails at any unit pair (refutes Theorem 29.1)	PASS
F-M28u.6	Simulation	Tier-3 PASS reverses to FAIL at extended trial count (50 000 $\rightarrow$ 500 000) on any V_4 channel (refutes Theorem 29.2)	PASS (legacy)
F-M28u.7	Mathematical	V_4 conductor identity $\log(3) + \log(11) \neq \log(33)$ at any precision (refutes Theorem 30.1)	PASS
F-M28u.8	External Dep.	PNT (Hadamard - de la Vallee Poussin 1896) is shown false (refutes Theorem 28.2)	PASS (standard)
F-M28u.9	External Dep.	Mordby-Rowlett 2024 is retracted or its $a_1(\text{equilateral}) = 1/3$ result shown incorrect (refutes W2' inheritance)	PASS
F-M28u.10	External Dep.	CCM 2025 [19] is retracted or its $D_{\log}(\lambda, N)$ construction shown incorrect (refutes Theorem 30.2 spectral bridge)	PASS
F-M28u.11	Mathematical	$F(\text{icosahedron}) \neq 20$ (refutes Theorem 31.4 multiplicity)	PASS
F-M28u.12	Mathematical	Lame eigenvalues do not match $m^2 + m n + n^2$ norm structure (refutes Theorem 31.4)	PASS
F-M28u.13	Anti-Overclaim	Theorem 28.4, Theorem 28.5, or Corollary 28.5a is interpreted or claimed as a proof of RH	PASS (NC-M28u.5, NC-M28u.6, NC-M28u.7)
F-M28u.14	Anti-Overclaim	Any §4-§9 result is found to introduce a new free parameter beyond LOCKED A, Q, K	PASS (audit per §3)
F-M28u.15	Mathematical	$T(T(z^*)) \neq z^*$ at any precision (refutes Corollary 28.5a Step 2)	PASS (1e-50)
F-M28u.16	Simulation	$ T'(z^*) \geq 1$ at higher precision (refutes attracting fixed point claim, Corollary 28.5a Step 1)	PASS

Table 11.1. ZS-M28 v1.0 falsification gates. All 16 gates currently PASS at the verification level. Layers covered: mathematical breakdown (F-M28u.1, .2, .7, .11, .12, .15), simulation/consistency (F-M28u.3, .4, .5, .6, .16), external dependency (F-M28u.8, .9, .10), anti-overclaim (F-M28u.13, .14).

§12. NON-CLAIMS (Inherited and Consolidated)

All non-claims of the four legacy drafts are inherited verbatim and re-tagged. The cumulative anti-overclaim posture of the Z-Spin RH program (NC-M22.1 through NC-M31.8) is preserved in full strength.

NC-M28u.1: This paper does NOT claim a proof of the Riemann Hypothesis or any GRH for $K = Q(\sqrt{-3}, \sqrt{-11})$. Inherited from NC-M22.1, NC-M23.1, NC-M28.5, NC-M29.1, NC-M30.2, NC-M31.3.

NC-M28u.2: This paper does NOT close W2 (V_4 Weil functional positivity). The corpus PROVEN no-go ADS-5 [25] and ADS-6 [25] are uncircumvented. Closure remains external Connes (2000) [12] / Burnol (1998-2004) [14-18] / CCM 2025 [19] under D4b.

NC-M28u.3: Theorem 30.1 is at HYPOTHESIS-strong status at the constant level only. The corresponding operator-level functor is OPEN (legacy O-M30.1').

NC-M28u.4: Theorem 30.2 is at HYPOTHESIS-strong status at the spectral data level only. It does NOT claim operator-level isomorphism or unitary equivalence between corpus $L_s^*(P)$ and CCM 2025 $D_{\log}(\lambda, N)$. Operator-level equivalence requires explicit form of $D_{\log}$ from CCM 2025 §3-§5 [19] and is OPEN (legacy O-M30.4').

NC-M28u.5: Theorem 28.4 (HYPOTHESIS-strong) does NOT prove RH. The connection $\log|D|^2 \sim -(2Q/D^*) \log|\zeta_P|$ holds for any s with $\sigma > 0$; it does not constrain Riemann zero locations to $\sigma = 1/2$.

NC-M28u.6: Theorem 28.5 (DERIVED-interpretation) does NOT prove RH. It establishes the structural isomorphism $\sigma = 1/2 \leftrightarrow j = 1/2$ at the Z_2 -involution level only. RH still requires that all non-trivial zeta-zeros actually lie on the critical line — OPEN (P1-P4 of ZS-QS [29]).

NC-M28u.7: Corollary 28.5a (HYPOTHESIS-strong) does NOT prove RH. The Mobius-trace reading specifies that Z-Spin and RH ride together (dong-seung) under the same Z_2 involution but do NOT share count-axis observables. The Riemann count of zeros is undefined on the Z-Spin side; the Z-Spin Mobius traversal has no Riemann analog.

NC-M28u.8: Theorem 31.4 (DERIVED-CANDIDATE) does NOT directly close W2. Multiplicity 20 of the $L(s, \chi_{-3})$ realization does not change L-function sign behavior. The discovery is structural: it identifies WHERE in Z-Spin the χ_{-3} channel lives, not HOW to close W2. Inherited from NC-M31.6, NC-M31.7.

NC-M28u.9: This paper does NOT extend the Langlands correspondence beyond abelian class field theory. K is a degree-4 abelian extension of Q ; the correspondence between its Galois representations and automorphic forms is fully PROVEN mathematics (class field theory). Inherited from NC-M23.4.

NC-M28u.10: This paper does NOT introduce any new free parameter beyond $A = 35/437$ (LOCKED, ZS-F2 [2]), $Q = 11$ (PROVEN, ZS-F5 [3]), K (PROVEN, ZS-M22 [25]). All inputs trace to LOCKED, PROVEN, DERIVED, or IMPORTED items in upstream corpus papers or external citations.

§13. OPEN Problems and Closure Paths

The unified release inherits all open problems of the legacy drafts and consolidates them under a single roster:

ID	Statement	Closure Path
O-M28u.1	Q-tower extension: does a sequence $Q_n = 11 k_n$ exist for which LOCATOR MAD $\rightarrow 0$?	Internal: scan Q in $\{11, 33, 55, \dots\}$; relate to dimension extension
O-M28u.2	Small-prime correction structure: does R_{small} encode arithmetic information beyond $\log p$ oscillations?	Internal: relate to Davenport-Heilbronn off-line zero detection
O-M28u.3	W1 closure to PROVEN: trace-norm convergence of $H_{\infty}^{\{\text{Yak}, J\}}$ (legacy O-M26.2)	External: trace-norm convergence theorem extending Yakoboylu 2024 [16]
O-M28u.4	W2 closure: V_4 Weil functional positivity via D4b conductor / parity correction	External: Connes (2000) [12] / Burnol (1998-2004) [14-18] / CCM 2025 [19]
O-M28u.5	Operator-level functor B_Z for Theorem 30.1 (legacy O-M30.1')	External: explicit form of CCM 2025 $D_{\log}^{\lambda, N}$ operator [19]
O-M28u.6	Operator-level isomorphism for Theorem 30.2 (legacy O-M30.4')	External: explicit construction of $D_{\log}^{\lambda, N}$ per CCM 2025 §3-§5
O-M28u.7	Selberg-icosahedron W2 path (legacy ZS-M31 §7.3, NEW)	Internal+External: Selberg trace formula on closed icosahedral surface, V_4 functional adapted to closed-geodesic length spectrum
O-M28u.8	Truncation-Dual Theorem promotion: χ_{-3} carrier on triangular face vs hexagonal face (legacy ZS-F0 §11.2)	Internal: explicit relation of pre-truncation Lame spectrum to truncated-icosahedron face spectrum
O-M28u.9	i-tetration $\rightarrow$ operator promotion: Koopman lift onto reproducing kernel Hilbert space (Mauroy-Mezic 2020, Boulle-Colbrook-Conradie 2025)	External: prime indexing supplied by external framework, not Z-Spin
O-M28u.10	Cohomology-level conductor decoration: $\alpha = 0$ limit unique to Kostant?	Internal: explore $\alpha \neq 0$ with alternative grading structures (legacy Result M29.4)

Table 13.1. Open problems consolidated from legacy ZS-M28-M31. Eight items inherited, two added (O-M28u.7 from legacy ZS-M31 §7.3, O-M28u.10 from legacy Result M29.4). Four are OPEN under external import; six admit internal Z-Spin closure paths.

§14. Conclusion

This paper consolidates four legacy drafts into a single bridge for the Z-Spin RH program. The bridge thesis is: Z-Spin and RH ride together (dong-seung) under the same Z_2 involution structure (Theorem 28.5), but the Riemann count of zeros has no internal observable on the Z-Spin side (Corollary 28.5a). Where the two sides ride together, we have PROVEN structural isomorphisms (the nine $1/2$ manifestations, Table 8.1). Where they diverge, we have walls (W1, W2, W3) that are not problems but maps of the divergence.

Theorem 28.1 closes wall W1 to DERIVED-CONDITIONAL on PNT alone, by establishing that $L_s(P)$ is diagonal in the computational basis. Theorem 28.4 (HYPOTHESIS-strong) explains why corpus PROVEN LOCATOR peaks coincide with Riemann zero heights. Theorem 29.1 establishes the V_4 multi-channel Triple Structure with Tier-3 anti-numerology PASS at 50 000 trials (Theorem 29.2).

Theorems 30.1-30.3 identify three external mathematical vehicles for D4b closure and establish constant-level conductor identification. Theorem 31.4 closes the geometric origin gap for the $\chi_{-3} V_4$ channel, identifying the 20 equilateral triangular faces of the pre-truncation icosahedron as the geometric carrier.

Theorem 28.5 (NEW, DERIVED-interpretation) and Corollary 28.5a (NEW, HYPOTHESIS-strong) are the two contributions of this unified release: a single-frame consolidation of corpus 1/2-manifestations under one Z_2 -involution reading, and a Mobius-trace reading of the RH count of zeros as a count-invariant traversal of the i-tetration fixed point z^* . Both are interpretive layers; neither proves RH (NC-M28u.6, NC-M28u.7).

What this paper achieves is not RH but its Z-Spin meaning. The Z-Spin framework now carries an explicit, anti-numerology-checked, falsification-gated reading of why the Riemann critical line is $\sigma = 1/2$ in dong-seung with the Z-sector spinor 4π closure. Whether RH itself is ultimately proven, refuted, or remains open, the Z-Spin corpus contribution to the question is now mapped as a single 30-test verified bridge.

Acknowledgements & Code Availability

This work was developed across 47+ exploratory rounds with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, external literature search, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

The companion verification suite `zs_m28_unified_verify_v1_0.py` is publicly available at github.com/KennyKang-git/zspin/papers/ZS-M28/. Dependencies: Python ≥ 3.9 , NumPy, SciPy, SymPy, mpmath (50-digit precision for z^* -related identities). Execution: `python3 zs_m28_unified_verify_v1_0.py`. Expected output: 30/30 PASS, exit code 0. The four legacy verify scripts (`zs_m28_verify_v1_0.py`, `zs_m29_verify_v1_0.py`, `zs_m30_verify_v1_0.py`, `zs_m31_verify_v1_0.py`) are preserved in github.com/KennyKang-git/zspin/papers/ZS-M28/legacy/ for transparency.

Appendix A. Cross-Reference: Legacy Theorem Codes

All legacy theorem codes from ZS-M28, ZS-M29, ZS-M30, ZS-M31 (May 2026 internal drafts) map into the unified ZS-M28 v1.0 (March 2026) as follows. Downstream papers should cite the unified codes.

Legacy Code	Statement (compressed)	Unified Code	Status
M28.1	$L_s(P)$ diagonal in computational basis	Theorem 28.1	PROVEN
M28.2	$M_P(s) \rightarrow I$ in trace-norm under PNT	Theorem 28.2	DERIVED-CONDITIONAL
M28.3	Dirichlet kernel exact identity	Theorem 28.3	PROVEN
M28.4	$\log D ^2 \sim -(2Q/D^*) \log \zeta_P $	Theorem 28.4	HYPOTHESIS-strong
M29.1	V_4 multi-channel Triple Structure	Theorem 29.1	DERIVED

Legacy Code	Statement (compressed)	Unified Code	Status
M29.2	V_4 Tier-3 Anti-Numerology PASS	Theorem 29.2	DERIVED
M29.3	Q-Scan Stability of LOCATOR	Theorem 29.3	DERIVED-CONDITIONAL
M29.4	Cohomology-level confirmation of NC-M27.4	Result 29.4	OBSERVATION
M30.1	Constant-Level Conductor Identification	Theorem 30.1	HYPOTHESIS-strong
M30.2	LOCATOR $\leftrightarrow$ D_log Spectral Bridge	Theorem 30.2	HYPOTHESIS-strong
M30.3	J seam $\leftrightarrow$ Burnol Grading Non-isomorphism	Theorem 30.3	PROVEN
M31.1 (T31)	V_4 analog of CC2021 Cor 2.3 (11/11 PASS)	Result 31.1	DERIVED-CONDITIONAL
M31.4 (T32)	Y-sector pre-truncation icosahedral face-wave Eisenstein carrier	Theorem 31.4	DERIVED-CANDIDATE
—	$\sigma = 1/2 \leftrightarrow j = 1/2$ dynamical equilibrium	Theorem 28.5 (NEW)	DERIVED-interpretation
—	RH count as Mobius trace of i-tetration	Corollary 28.5a (NEW)	HYPOTHESIS-strong

Table A.1. Cross-reference between legacy ZS-M28 / M29 / M30 / M31 (May 2026 drafts) and unified ZS-M28 v1.0 (March 2026).

References

Internal (Z-Spin Cosmology v2.0 corpus)

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Version History

v1.0 (March 2026, unified): Initial public release of ZS-M28 v1.0 (unified). Theorems 28.1 (PROVEN), 28.2 (DERIVED-CONDITIONAL on PNT), 28.3 (PROVEN), 28.4 (HYPOTHESIS-strong), 29.1 (DERIVED), 29.2 (DERIVED), 29.3 (DERIVED-CONDITIONAL), 30.1 (HYPOTHESIS-strong), 30.2 (HYPOTHESIS-strong), 30.3 (PROVEN), 31.4 (DERIVED-CANDIDATE), 28.5 (NEW, DERIVED-interpretation), Corollary 28.5a (NEW, HYPOTHESIS-strong). Verification suite 30/30 PASS at mpmath 50-digit precision (numerical) plus algebraic exactness (diagonal structure, Schur orthogonality, V_4 closure). Falsification gates F-M28u.1 through F-M28u.16 registered, all PASS. Open problems O-M28u.1 through O-M28u.10 registered. Non-claims NC-M28u.1 through NC-M28u.10

registered. Zero new free parameters; $A = 35/437$, $Q = 11$, $K = Q(\sqrt{-3}, \sqrt{-11})$ LOCKED throughout. SUPERSEDES legacy ZS-M28 (May 2026 v1.0), ZS-M29 (May 2026 v1.0), ZS-M30 (May 2026 v1.0), ZS-M31 (March 2026 v1.0 augmented). Consolidated from internal Z-Spin Collaboration research notes through the four legacy drafts and Kenny Kang's March 2026 unified-bridge synthesis directive.

(Internal version history: legacy v3.1.0 -> unified v1.0. Legacy ZS-M28 v1.0 (May 2026) ran 28 verification tests; legacy ZS-M29 v1.0 (May 2026) ran 21; legacy ZS-M30 v1.0 (May 2026) ran 22; legacy ZS-M31 v1.0 (March 2026 augmented) ran 22. Unified suite consolidates these into 30 tests after removing redundant locked-input re-checks and helper-only stubs absorbed by upstream PROVEN tags.)